

JEE MAIN REVISION MASTER: COMPOUNDS CONTAINING NITROGEN

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 229 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Diazonium Salts & Carbylamine Test	120 PYQs	Diazonium & Tests
Very High	Basicity of Amines & Gabriel Phthalimide Synthesis	111 PYQs	Amine Basics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Diazonium Salts & Carbylamine Test	$\text{Diazotization: } \text{Ar-NH}_2 + \text{NaNO}_2 + 2\text{HCl} \rightarrow \text{Ar-N}_2^+ \text{Cl}^- + \text{NaCl} + 2\text{H}_2\text{O}$
Basicity of Amines & Gabriel Phthalimide Synthesis	$\text{Basicity in water: } (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N} > (\text{CH}_3)_3\text{N}^+ > (\text{CH}_3)_3\text{N} > (\text{CH}_3)_3\text{N}^+ > (\text{CH}_3)_3\text{N} > (\text{CH}_3)_3\text{N}^+$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Diazonium Salts & Carbylamine Test (Very High Yield — 120 PYQs)

Core Principle: Preparation of benzene diazonium chloride (0-5°C), Sandmeyer/Gattermann halogenation, azo coupling dyes, and Hinsberg separation.

■ **Caution / JEE Pitfall:** Carbylamine test gives foul-smelling isocyanide ONLY with primary amines (1°); secondary and tertiary amines give NO reaction.

■ **Speed Trick:** Hinsberg reagent (C₆H₅SO₂Cl): 1° amine gives product soluble in KOH; 2° amine gives product insoluble in KOH; 3° amine does not react.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following compounds according to increasing order of boiling points. $n\text{-C}_4\text{H}_9\text{OH}(A), n\text{-C}_4\text{H}_9\text{NH}_2(B), n\text{-C}_4\text{H}_{10}(C)$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Product C of the following reaction sequence will be..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider the following reactions giving major product. Identify the correct reaction...." [Ans: N/A]

3.2 Basicity of Amines & Gabriel Phthalimide Synthesis (Very High Yield — 111 PYQs)

Core Principle: Interplay of inductive, steric, and solvation factors on amine basicity in gas vs aqueous phases, and Hoffmann bromamide degradation.

■ **Caution / JEE Pitfall:** In gas phase, basicity order is purely inductive: $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$. In aqueous solution, solvation and steric factors disrupt this order.

■ **Speed Trick:** Gabriel phthalimide synthesis yields PURE aliphatic primary amines (1°); aromatic amines cannot be prepared because aryl halides do not undergo SN_2 .

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Which statements are True? A. In Hoffmann bromamide degradation, 4 moles of NaOH and 2 moles of Br_2 are consumed per mole of an amide B..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#2 (MCQ):** "Which of the following amino acid will give violet coloured complex with neutral ferric chloride solution?..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#2 (MCQ):** "Given below are two statements : Statement I : Heating benzamide with bromine in an ethanolic solution of sodium hydroxide will give benzyla..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Arrange the following compounds according to increasing order of boiling points.

$\text{n-C}_4\text{H}_9\text{OH}$ (A), $\text{n-C}_4\text{H}_9\text{NH}_2$ (B), $\text{n-C}_{10}\text{H}_{22}$ (C) and $\text{C}_2\text{H}_5\text{NHC}_2\text{H}_5$ (D).

- (A) $\text{C} < \text{B} < \text{A} < \text{D}$
- (B) $\text{D} < \text{C} < \text{B} < \text{A}$
- (C) $\text{C} < \text{D} < \text{B} < \text{A}$
- (D) $\text{D} < \text{B} < \text{A} < \text{C}$

Correct Answer: None

Solution Analysis: (A) $\text{n-C}_4\text{H}_9\text{OH}$ (an alcohol) forms strong intermolecular hydrogen bonding between O-H groups of different molecules. Because of this strong attraction, more heat is needed to separate the molecules, so its boiling point is the highest. (B) $\text{n-C}_4\text{H}_9\text{NH}_2$ (a primary amine) also shows intermolecular hydrogen bonding due to the presence of N-H bonds. But nitrogen is less electronegative than oxygen, so N-H hydrogen bonding is weaker than O-H hydrogen bonding. Hence, its boiling point is lower than the alcohol. (C) $\text{n-C}_{10}\text{H}_{22}$ is a non-polar hydrocarbon. It cannot form hydro

Benchmark Problem 2: JEE Main 2026 (Online) 24th January Evening Shift (Q#2)

Given below are two statements : Statement I : The dipole moment of R-CN is greater than R-NC and R-NC can Statement II : R-CN hydrolyses under acidic medium to produce a compound which on treatment with SOCl_2 , followed by the addition of NH_3 gives another compound (x). This compound (x) on treatment with NaOCl/NaOH gives a product, that on treatment with $\text{CHCl}_3/\text{KOH}/\Delta$ produces R-NC. In the light of the above statements, choose the correct answer from the options given below

- (A) Both Statement I and Statement II are false
- (B) Statement I is false but Statement II is true
- (C) Statement I is true but Statement II is false
- (D) Both Statement I and Statement II are true

Correct Answer: None

Solution Analysis: Statement I : False $\text{R-NC} \rightarrow \text{R-NH}_2 + \text{HCOOH}$ Statement II : True $\text{R-CN} \xrightarrow{\text{H}^+} \text{R-COOH} \xrightarrow{\text{SOCl}_2} \text{R-COCl} \xrightarrow{\text{NH}_3} \text{R-CONH}_2 \xrightarrow{\text{CHCl}_3/\text{KOH}/\Delta} \text{R-NC}$